

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi – 590018



"VTU Internship Report"

ON

"Android App Development using Generative AI"

Submitted in partial fulfilment of the requirements for the award of the degree in the 8th semester.

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

BY

KSHITIZ KANNOJIA

1ME22CS066

Under the Guidance of

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External Guide

Program Coordinator,

MindMatrix



MS ENGINEERING COLLEGE

Approved By AICTE New Delhi, Affiliated to VTU, Belagavi, Karnataka

NAAC Accredited and An ISO 9001:2015 Certified Institution

Department of Computer Science and Engineering

Academic Year 2025 – 2026



MS ENGINEERING COLLEGE

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Navarathna Agrahara, Sadhalli Post, Off Bengaluru International Airport, Bengaluru – 562110



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



CERTIFICATE

This is to certify that the Internship Report entitled "**Android App Development using Generative AI – Kreed-Prerana Scout**" is a Bonafide work carried out by **Kshitiz Kannoja (1ME22CS066)** in partial fulfilment for the award of **Bachelor of Engineering in Computer Science and Engineering of Visvesvaraya Technological University, Belagavi**, during the year **2025–2026**. It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report. This internship report has been approved as it satisfies the academic requirements in respect of internship work prescribed for the Bachelor of Engineering Degree.

Signature of Guide

Mr. Venkatesha Nayak

Assistant Professor

Dept. of ISE

Signature of Coordinator

Ms. Harshitha S P

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Dept. of CSE

Signature of HOD

Dr. Malatesh S H

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Dept. of CSE

Signature of Principal

Dr. N Ranapratap Reddy

Principal of MSEC

1. Internal Examiner Name: _____ Signature _____

2. External Examiner Name: _____ Signature _____

Exam Date:

INTERNSHIP COMPLETION LETTER

MindMatrix, an MoU partner of Visvesvaraya Technological University (VTU), has successfully conducted the internship program titled **"Android App Development using Generative AI"**, announced on the VTU internship portal.

The internship was conducted in **online** mode from **2nd February 2026 to 18th May 2026**, in accordance with **industry-standard processes and practices followed** by MindMatrix.

During the internship, the student was exposed to industry-aligned learning and hands-on practice in the following areas:

1. Android application development using **Kotlin** and **Jetpack Compose**
2. Hands-on experience with **Android Studio**, **Google Cloud Labs**, and **Google AI Studio** to explore **GenAI capabilities** in application development
3. **UI/UX design**, application prototyping, feature development, followed by **debugging and iterative enhancements**
4. **Project-based learning** involving **Firebase integration**, testing, and standard **Android development workflows**

The following student was enrolled in the internship program through the VTU internship portal, and is subject to verification and validation by the respective college.

Name: Kshitiz Kannoja
USN: 1ME22CS066
College Name: M S Engineering College

The student's performance during the internship was evaluated based on **continuous participation, task completion, and overall engagement on the online platform MindMatrix.io**. Detailed performance metrics as per internship program coordinators' judgement is provided in the **attached annexure**.

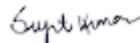
The intern's overall performance during the internship was assessed based on the prescribed evaluation criteria and rated as **EXCELLENT** / Very Good / Good / Satisfactory / Needs Improvement

This evaluation is for internship assessment only and does not carry direct academic marks.

The internship completion letter is issued at the student's request, based on records available with MindMatrix, without any risk or liability to the organization or its authorized signatories.

We wish the student the very best in their future academic and professional endeavors.

Disclaimer: For official use by authorized Internship Coordinators of the respective college only.


Sujit Kumar
Founder and Director,
[MindMatrix.io](https://mindmatrix.io)
CL Infotech Pvt. Ltd.

Date: 17th May, 2026
Letter ID: Mx26INT01883

To verify, visit <https://lms.mindmatrix.io/>

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DECLARATION

I, **Kshitiz Kannoja (1ME22CS066)**, Students of 8th Semester B.E Computer Science and Engineering, **M S Engineering College**, Bangalore, Declare that this Internship work Entitled “**Android App Development Using Gen AI**” is the original work carried by me under the guidance and supervision provided during the internship period in the partial fulfilment of the course requirements for the award of the Degree in **Bachelor of Engineering in Computer Science and Engineering** by **Visvesvaraya Technological University, Belagavi** during the year 2025-2026.

SIGNATURE OF THE STUDENT

ACKNOWLEDGEMENT

I would like to express my deep sense of gratitude to our Principal **Dr. N Ranapratap Reddy**, M S Engineering College, Bengaluru for his inspiration and academic support by providing good facilities to complete the internship.

I extend special in-depth, heartfelt, and sincere gratitude to **Dr. Malatesh S H**, Head of CSE Department, for his wholehearted support in completion of this internship.

I would like to express my heartfelt gratitude to **Mr. Tirumal Mutalikdesai**, External Internship Guide at MINDMATRIX, for giving me the opportunity, guidance, and support to complete my internship.

My sincere thanks to my internship coordinator **Ms. Harshitha S P**, Assistant Professor, Dept. of CSE, for guiding and giving me timely advice and suggestions in the successful completion of the internship.

I am highly indebted to my Internal Internship Guide **Mr. Venkatesha Nayak**, Assistant Professor, Dept. of ISE, for guiding and giving me timely advice and suggestions in the successful completion of the internship.

I thank all the staff members for supporting me in completion of the internship. Also, I am thankful to my parents and all my dear friends who have directly or indirectly helped me.

Finally, I express gratitude to all those who knowingly or unknowingly helped me in successful completion of this internship.

KSHITIZ KANNOJIA
(1ME22CS066)

ABSTRACT

This report details a comprehensive 12-week project-based internship completed at MindMatrix (February 3, 2026 – April 28, 2026), focusing on Android App Development. The internship followed a structured three-phase learning model covering foundational Android concepts, core feature development, and capstone project delivery.

The technical phase involved creating high-performance Android applications using Java, XML Layouts with Material Design Components, Room Database for offline persistence, and MPAndroidChart for data visualization. Emphasis was placed on MVVM architecture, background threading with Executors, and reactive UI updates using LiveData.

The capstone project delivered during Phase 3 of the internship was Kreed-Prerana Scout — a grassroots sports talent tracker Android application designed for rural school physical education teachers. The app enables PE teachers to record student athlete profiles, log trial performances with a high-precision stopwatch (10ms accuracy), perform batch entry for 30 students simultaneously, view leaderboard rankings, and track performance improvement through Talent Curve graphs. Automatic “District Level Ready” badges are awarded when students cross preset sport-specific benchmarks across 13 events.

Beyond technical skills, the internship cultivated the Campus to Corporate (C2C) framework, instilling professional ethics, structured problem-solving using DMAIC methodology, and effective communication through the PREP method.

Keywords: Android Development, Java, XML Layouts, Room Database, MPAndroidChart, Material Design, MVVM Architecture, Sports Talent Tracking, Offline-First, LiveData, RecyclerView, Stopwatch Timer, Batch Entry, Leaderboard

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CHAPTER 1

ABOUT THE ORGANIZATION

The following section provides a detailed overview of the organization where the internship was conducted, highlighting its structure, mission, and operational focus.

1.1 Profile of the Organization (MindMatrix)

ASPECT	DETAILS
Organization Name	MindMatrix (Ed-Tech and Software Ecosystem Company)
Nature of Organization	Outcome-driven AI-fluency and skill development training platform for engineering students
Founded By	Mr. Sujit Kumar
Headquarters	MindMatrix, #651, Kiran Arcade, 4th Floor, 13th Cross, 27th Main Road, 1st Sector, HSR Layout, Bengaluru – 560102 (Online / Virtual Operations across India)
India Operations	Bengaluru, Karnataka – Primary development and internship hub
Internship Track	Android App Development using Generative AI
Internship Type	Online / Remote Internship
Internship Duration	12 Weeks – February 3, 2026 to April 28, 2026
Program Coordinator	Mr. Tirumal Mutalikdesai
Industry Mentors	Mr. Ramakrishna Rama, Mr. Aravind Warriar

1.2 Vision and Mission

ASPECT	DETAILS
Vision Statement	To empower organizations and engineering students with intelligent, AI-driven sales enablement and learning solutions that drive measurable business outcomes — producing industry-ready engineers who can design, build, and deploy intelligent software systems with professional ethics and structured problem-solving skills.
Mission Statement	• Deliver innovative software products powered by AI and automation. • Enable students and professionals through structured, industry-aligned internship programs.

	• Bridge the gap between academic knowledge and real-world software engineering. • Transition students from theoretical learners to professional 'System Builders'. • Cultivate AI-fluency and modern engineering practices. • Instil the Campus to Corporate (C2C) mindset across the talent pipeline.
Core Values	• Innovation: Encouraging creative and practical AI-powered solutions. • Integrity: Maintaining ethical standards in product development. • Curiosity: Fostering continuous learning and questioning. • Collaboration: Strengthening academic–industry partnerships. • Excellence: Delivering high-quality software that meets user expectations.

1.3 Key Services & Domains Served

ASPECT	DETAILS
Core Products	Sales Enablement Platform, AI-powered content delivery, Partner Management Systems
Technical Training	Android App Development, Generative AI Integration, Kotlin, Jetpack Compose, Room Database, MVVM Architecture, API integration
Project-Based Learning	Students build production-ready applications through structured Agile sprints over the internship duration
Mentorship	Industry mentors provide guidance on software engineering practices, DMAIC, 5 Whys, and PREP communication method
C2C Framework	Campus to Corporate: professional ethics, leadership values, structured communication, career readiness
Performance Tools	Google Antigravity tool for Critical Path optimization; Git for version control; Discord for Agile standups
Technology Focus	Artificial Intelligence, Android Development, Cloud Services, API Integration
Key Domains Served	EdTech, SaaS, Enterprise Sales Tools, Mobile Applications, AI-Assisted Software Development

CHAPTER 2

OBJECTIVES OF THE INTERNSHIP

The primary objective of this internship program at MindMatrix was to bridge the divide between theoretical academic knowledge and practical industry-level software development. Through a structured, project-based curriculum, the program aimed to equip interns with production-grade Android development skills, culminating in the delivery of a real-world application — Kreeda-Prerana Scout.

2.1 Technical Objectives: Android & GenAI Engineering

2.1.1 Mastery of Android Architecture

Develop a strong foundation in modern Android development practices including:

- Understanding MVVM Architecture with Activity controllers for separation of concerns.
- Implementing XML Layouts with Material Design Components for structured, responsive UI development.
- Managing background operations using `Executors.newSingleThreadExecutor()` and `Handler` with `Runnable` callbacks.
- Using `RecyclerView` with custom adapters for efficient list rendering across multiple screens.

2.1.2 Offline-First Data Persistence

Integrate offline-first persistence into the Android application:

- Integrating Room Database for fully offline-first persistent storage of all athlete profiles and trial records.
- Implementing Room Database with DAO interfaces, LiveData queries, and `@ForeignKey` CASCADE relationships.
- Building a batch entry flow that records performance data for 30 students in a single session.
- Designing entity relationships between Athlete and TrialRecord tables with referential integrity.

2.1.3 Data Visualization & Performance Tracking

Integrate charting and performance analysis into the mobile application:

- Learn `MPAndroidChart` library integration for rendering line graphs with cubic Bezier interpolation.
- Integrate the `MPAndroidChart v3.1.0` library to display Talent Curve graphs showing performance trends over time.
- Implement custom `ValueFormatters` for date-based X-axis and score-based Y-axis labeling.
- Add `LimitLine` objects to display district-level benchmark thresholds directly on performance graphs.

2.1.4 Performance Optimization

Apply industry-grade performance engineering to mobile applications:

- Implement a high-precision stopwatch using `SystemClock.elapsedRealtime()` with 10ms Handler callbacks.
- Optimize RecyclerView rendering with ViewHolder pattern and efficient adapter data binding.
- Design BenchmarkUtils with dual comparison logic for timed events (lower-is-better) and distance events (higher-is-better).
- Implement automatic badge detection across 13 sport categories with preset district-level thresholds.

2.2 Professional Objectives

2.2.1 Version Control & Collaboration

- Master Git and GitHub workflows including branching and commit best practices.
- Securely manage sensitive credentials using `.gitignore` and `local.properties`.
- Share project source code professionally with internship coordinators via GitHub.

2.2.2 Problem Solving & Debugging

- Use Android Studio's debugger and Logcat to isolate runtime issues.
- Resolve Gradle build errors and dependency conflicts systematically.
- Test app functionality across different Android API levels.

2.2.3 Technical Documentation

- Prepare a comprehensive internship report documenting design decisions.
- Write clear README documentation for GitHub repository.
- Present system architecture and project outcomes to internship coordinators.

CHAPTER 3

LEARNING EXPERIENCES

This chapter details the theoretical knowledge acquired and the practical implementation milestones achieved during the 12-week internship at MindMatrix, structured into Foundation, Core Features, and Final Delivery phases.

3.1 Theoretical Foundation

Before commencing development, extensive time was dedicated to understanding the underlying concepts of Android development, offline-first persistence with Room, and Generative AI.

3.1.1 Android Architecture & Java Fundamentals

- Android Lifecycle: Activity lifecycle management; handling configuration changes and back-stack navigation across 7 screens.
- Java Threading: `Executors.newSingleThreadExecutor()` for background database operations, `Handler` with `Runnable` for timer callbacks.
- Java Language Features: Interfaces, abstract classes, `HashMap` collections, enum types, anonymous inner classes.
- Material Design: `MaterialComponents` theme, `MaterialCardView`, `FloatingActionButton`, `Spinner`, `MaterialToolbar`.

3.1.2 XML Layouts & MVVM Pattern

- XML Layouts: Building responsive UI with `ConstraintLayout`, `LinearLayout`, and `RecyclerView` components.
- State Management: Using `LiveData` for reactive database observation and UI updates.
- Data Flow: Activity captures user input, delegates to DAO/database layer, observes `LiveData` for results.
- MVVM Architecture: View Layer (XML + Activities), ViewModel/Controller Layer (Activities + `BenchmarkUtils`), Model Layer (Room DB).

3.1.3 Offline Persistence & Chart Integration

- Room Database: Local SQLite abstraction with `@Entity`, `@Dao`, `@Database` annotations for offline-first storage.
- LiveData Queries: Reactive query results that automatically update UI when database changes occur.
- Foreign Key Relationships: `@ForeignKey` CASCADE delete between `Athlete` and `TrialRecord` for referential integrity.
- `MPAndroidChart`: `LineChart` with cubic Bezier, custom `ValueFormatters`, `LimitLine` for benchmark thresholds.
- `BenchmarkUtils`: Utility class with 13 sport-specific benchmarks using `TIMED_BENCHMARKS` and `DISTANCE_BENCHMARKS` HashMaps.

3.2 Weekly Progress Report

The following tables summarize the chronological progression of tasks, challenges, and outcomes over the internship duration.

3.2.1 Phase 1: Foundation & Setup (Weeks 1–4)

Focus: Project setup, Android fundamentals, Room DB schema design, and Athlete Profile UI.

Week	Focus Area	Key Activities & Learnings
Week 1 03/02/2026 – 09/02/2026	Orientation & Android Basics	- Studied MindMatrix C2C framework and Internship Programme Handbook. - Set up Android Studio, created first Java project. - Studied Activity lifecycle and Android project structure. - Learned XML layout basics: LinearLayout, ConstraintLayout. - Built first Material Design screen with MaterialToolbar.
Week 2 10/02/2026 – 16/02/2026	Java & RecyclerView	- Practiced Java collections, interfaces, and abstract classes. - Built RecyclerView with custom ViewHolder and Adapter. - Implemented MaterialCardView for list item design. - Learned Spinner widget for sport category selection. - Created splash screen with app branding.
Week 3 17/02/2026 – 23/02/2026	Room DB & Persistence	- Set up Room Database dependencies in build.gradle. - Designed Athlete entity with @Entity and @PrimaryKey annotations. - Designed TrialRecord entity with @ForeignKey CASCADE. - Created AthleteDao with insert, update, delete, and query methods. - Created TrialRecordDao with LiveData return types.
Week 4 24/02/2026 – 02/03/2026	MVVM Architecture & Database	- Designed MVVM architecture: View → Controller → Model layers. - Implemented AppDatabase singleton with Room.databaseBuilder. - Built Athlete list screen with RecyclerView and FAB. - Implemented Add Athlete screen with form validation. - Connected Activities to Room DB via Executors for background writes.

3.2.2 Phase 2: Core Features & Alerts (Weeks 5–8)

Focus: Trial Logger stopwatch, Batch Entry for 30 students, Leaderboard ranking, and Benchmark detection.

Week	Focus Area	Key Activities & Learnings
Week 5 03/03/2026 – 09/03/2026	Trial Logger & Stopwatch	• Implemented Trial Logger screen with sport event Spinner. • Built high-precision stopwatch using <code>SystemClock.elapsedRealtime()</code>. • Created Handler with 10ms Runnable callbacks for 2-decimal accuracy. • Connected stopwatch output to TrialRecord entity. • Implemented save-to-database flow with Executors.
Week 6 10/03/2026 – 16/03/2026	Batch Entry System	• Designed Batch Entry screen for 30-student simultaneous input. • Built RecyclerView with EditText rows for score entry. • Implemented HashMap-based data holder to prevent scroll-recycling data loss. • Added sport event selector and bulk save functionality. • Tested batch save with 30 records in a single transaction.
Week 7 17/03/2026 – 23/03/2026	Leaderboard & Ranking	• Implemented Leaderboard screen with sport category filter. • Built sorting algorithm: ascending for timed events, descending for distance. • Designed rank-based card backgrounds: gold, silver, bronze. • Connected leaderboard to Room DB with LiveData observation. • Added athlete name, school, and best score display per rank.
Week 8 24/03/2026 – 30/03/2026	Benchmark Detection & Badges	• Created BenchmarkUtils utility class with 13 sport benchmarks. • Implemented dual comparison: TIMED_BENCHMARKS (lower-is-better) and DISTANCE_BENCHMARKS (higher-is-better). • Built automatic District Level Ready badge detection. • Added badge display on Athlete Detail screen. • Tested benchmark logic across all 13 sport categories.

3.2.3 Phase 3: AI Integration & Delivery (Weeks 9–12)

Focus: MPAndroidChart integration, Talent Curve visualization, performance optimization, testing, and final delivery.

Week	Focus Area	Key Activities & Learnings
Week 9 31/03/2026 – 06/04/2026	MPAndroidChart Integration	• Integrated MPAndroidChart v3.1.0 library in build.gradle. • Built Talent Curve screen with LineChart for performance trends. • Implemented cubic Bezier interpolation for smooth curve rendering. • Created custom ValueFormatters for date X-axis and score Y-axis. • Added LimitLine for district-level benchmark visualization.
Week 10 07/04/2026 – 13/04/2026	Performance Optimization	• Optimized RecyclerView adapter with ViewHolder pattern. • Ensured all database operations run on background threads. • Tested batch entry with 30 students for performance validation. • Verified stopwatch accuracy across multiple devices. • Optimized Room DB queries with proper indexing.
Week 11 14/04/2026 – 20/04/2026	Testing & QA	• Tested all 7 screens for navigation and back-stack correctness. • Verified benchmark detection across all 13 sport events. • Tested Talent Curve graph with multiple trial records. • Validated batch entry data persistence after scroll recycling. • Fixed edge cases in timer precision and leaderboard sorting.
Week 12 21/04/2026 – 28/04/2026	Final Delivery	• Polished all UI screens with consistent Material Design styling. • Created README documentation for GitHub repository. • Final GitHub push with complete source code. • Prepared technical documentation and internship report. • Delivered final APK and project presentation to coordinators.

CHAPTER 4

TASKS PERFORMED

STUDENT'S DAY-WISE DIARY

The internship was structured over 12 weeks with specific objectives and deliverables each week. The following day-wise diary documents the activities performed throughout the internship period at MindMatrix.

Week – 1 (03/02/2026 – 09/02/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	03/02/2026	9:00 AM	12:00 PM	Orientation: MindMatrix C2C framework & Internship Programme Handbook	Consistently delivered tasks for week 1, with initiative and attention to detail.
2	04/02/2026	9:00 AM	12:00 PM	Android Studio setup & first Java project with Gradle sync	Consistently delivered tasks for week 1.
3	05/02/2026	9:00 AM	12:00 PM	Java basics review: OOP, interfaces, collections, HashMap	Consistently delivered tasks for week 1.
4	06/02/2026	9:00 AM	12:00 PM	Android Activity lifecycle & Intent navigation between screens	Consistently delivered tasks for week 1.
5	07/02/2026	9:00 AM	12:00 PM	XML layout fundamentals: LinearLayout, ConstraintLayout, dp/sp units	Consistently delivered tasks for week 1.
6	09/02/2026	9:00 AM	12:00 PM	Project brief: Kreedha-Prerana Scout requirements & feature planning	Consistently delivered tasks for week 1.

Mr. Tirumal Mutalikdesai – Industry Supervisor, MindMatrix

Ms. Deepthi – Internal Guide | Ms. Harshitha S P – Internal Coordinator

Week – 2 (10/02/2026 – 16/02/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	10/02/2026	9:00 AM	12:00 PM	Java abstract classes & interface implementation patterns	Consistently delivered tasks for week 2.
2	11/02/2026	9:00 AM	12:00 PM	RecyclerView fundamentals: Adapter, ViewHolder, LayoutManager	Consistently delivered tasks for week 2.
3	12/02/2026	9:00 AM	12:00 PM	Reusable XML layouts: MaterialCardView, custom list items	Consistently delivered tasks for week 2.

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
4	13/02/2026	9:00 AM	12:00 PM	Spinner widget for dropdown selection & event handling	Consistently delivered tasks for week 2.
5	14/02/2026	9:00 AM	12:00 PM	Splash screen design with Material Design branding	Consistently delivered tasks for week 2.
6	16/02/2026	9:00 AM	12:00 PM	Home Dashboard screen layout	Consistently delivered tasks for week 2.

Week – 3 (17/02/2026 – 23/02/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	17/02/2026	9:00 AM	12:00 PM	Room Database setup & build.gradle dependency configuration	Consistently delivered tasks for week 3.
2	18/02/2026	9:00 AM	12:00 PM	Athlete entity: @Entity, @PrimaryKey(autoGenerate), fields design	Consistently delivered tasks for week 3.
3	19/02/2026	9:00 AM	12:00 PM	TrialRecord entity: @ForeignKey CASCADE, sport event enum fields	Consistently delivered tasks for week 3.
4	20/02/2026	9:00 AM	12:00 PM	AthleteDao: insertAthlete, getAllAthletes, getAthleteById with LiveData	Consistently delivered tasks for week 3.
5	21/02/2026	9:00 AM	12:00 PM	TrialRecordDao: insertTrial, getTrialsForAthlete, getBestScore queries	Consistently delivered tasks for week 3.
6	23/02/2026	9:00 AM	12:00 PM	End-to-end Room DB testing on emulator	Consistently delivered tasks for week 3.

Week – 4 (24/02/2026 – 02/03/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	24/02/2026	9:00 AM	12:00 PM	MVVM architecture design: View, ViewModel/Controller, Model layers	Consistently delivered tasks for week 4.
2	25/02/2026	9:00 AM	12:00 PM	AppDatabase singleton with Room.databaseBuilder & fallbackToDestructiveMigration	Consistently delivered tasks for week 4.
3	26/02/2026	9:00 AM	12:00 PM	Room DB schema finalization: Athlete + TrialRecord with FK relationship	Consistently delivered tasks for week 4.

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
4	27/02/2026	9:00 AM	12:00 PM	LiveData observation pattern for reactive UI updates	Consistently delivered tasks for week 4.
5	28/02/2026	9:00 AM	12:00 PM	MainActivity: RecyclerView athlete list with FAB for adding new athletes	Consistently delivered tasks for week 4.
6	02/03/2026	9:00 AM	12:00 PM	LiveData observation pattern wired to RecyclerView adapter	Consistently delivered tasks for week 4.

Week – 5 (03/03/2026 – 09/03/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	03/03/2026	9:00 AM	12:00 PM	TrialLoggerActivity: sport event Spinner & score input field	Consistently delivered tasks for week 5.
2	04/03/2026	9:00 AM	12:00 PM	SystemClock.elapsedRealtime() stopwatch implementation	Consistently delivered tasks for week 5.
3	05/03/2026	9:00 AM	12:00 PM	Handler with 10ms Runnable callback for 2-decimal precision	Consistently delivered tasks for week 5.
4	06/03/2026	9:00 AM	12:00 PM	Stopwatch UI: start/stop/reset buttons with elapsed time display	Consistently delivered tasks for week 5.
5	07/03/2026	9:00 AM	12:00 PM	Save trial record to Room DB via Executors background thread	Consistently delivered tasks for week 5.
6	09/03/2026	9:00 AM	12:00 PM	Dashboard UI polish & loading states	Consistently delivered tasks for week 5.

Week – 6 (10/03/2026 – 16/03/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	10/03/2026	9:00 AM	12:00 PM	Batch Entry screen wireframe: 30-row RecyclerView with EditText	Consistently delivered tasks for week 6.
2	11/03/2026	9:00 AM	12:00 PM	BatchEntryAdapter: custom ViewHolder with score input per student	Consistently delivered tasks for week 6.
3	12/03/2026	9:00 AM	12:00 PM	HashMap data holder to persist values during RecyclerView scroll recycling	Consistently delivered tasks for week 6.
4	13/03/2026	9:00 AM	12:00 PM	Sport event selector & bulk save: insert 30 TrialRecords in one transaction	Consistently delivered tasks for week 6.

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
5	14/03/2026	9:00 AM	12:00 PM	Executors-based batch insert with progress feedback	Consistently delivered tasks for week 6.
6	16/03/2026	9:00 AM	12:00 PM	Quick Entry: end-to-end 5-second flow testing	Consistently delivered tasks for week 6.

Week – 7 (17/03/2026 – 23/03/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	17/03/2026	9:00 AM	12:00 PM	LeaderboardActivity: sport category Spinner filter	Consistently delivered tasks for week 7.
2	18/03/2026	9:00 AM	12:00 PM	Sorting algorithm: ascending for timed events, descending for distance	Consistently delivered tasks for week 7.
3	19/03/2026	9:00 AM	12:00 PM	Rank-based UI: gold/silver/bronze card backgrounds with drawable shapes	Consistently delivered tasks for week 7.
4	20/03/2026	9:00 AM	12:00 PM	RecyclerView leaderboard adapter with rank, name, score display	Consistently delivered tasks for week 7.
5	21/03/2026	9:00 AM	12:00 PM	LiveData query: getBestScoresBySport() with Room DB	Consistently delivered tasks for week 7.
6	23/03/2026	9:00 AM	12:00 PM	Customer search edge-case testing (empty query, no match)	Consistently delivered tasks for week 7.

Week – 8 (24/03/2026 – 30/03/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	24/03/2026	9:00 AM	12:00 PM	BenchmarkUtils: TIMED_BENCHMARKS HashMap (100m < 14s, 200m < 28s, etc.)	Consistently delivered tasks for week 8.
2	25/03/2026	9:00 AM	12:00 PM	BenchmarkUtils: DISTANCE_BENCHMARKS HashMap (Long Jump > 4.5m, etc.)	Consistently delivered tasks for week 8.
3	26/03/2026	9:00 AM	12:00 PM	Dual comparison logic: lower-is-better for time, higher-is-better for distance	Consistently delivered tasks for week 8.
4	27/03/2026	9:00 AM	12:00 PM	Automatic District Level Ready badge on AthleteDetailActivity	Consistently delivered tasks for week 8.

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
5	28/03/2026	9:00 AM	12:00 PM	Badge detection tested across all 13 sport event categories	Consistently delivered tasks for week 8.
6	30/03/2026	9:00 AM	12:00 PM	End-to-end flow testing: Trial Logger, Batch Entry, Leaderboard	Consistently delivered tasks for week 8.

Week – 9 (31/03/2026 – 06/04/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	31/03/2026	9:00 AM	12:00 PM	MPAndroidChart v3.1.0 library integration in build.gradle	Consistently delivered tasks for week 9.
2	01/04/2026	9:00 AM	12:00 PM	TalentCurveActivity: LineChart setup with LineDataSet configuration	Consistently delivered tasks for week 9.
3	02/04/2026	9:00 AM	12:00 PM	Cubic Bezier mode enabled for smooth performance curve rendering	Consistently delivered tasks for week 9.
4	03/04/2026	9:00 AM	12:00 PM	Custom ValueFormatter for date-based X-axis labeling	Consistently delivered tasks for week 9.
5	04/04/2026	9:00 AM	12:00 PM	LimitLine added for district benchmark threshold visualization	Consistently delivered tasks for week 9.
6	06/04/2026	9:00 AM	12:00 PM	Prompt iteration for tone, length, and accuracy	Consistently delivered tasks for week 9.

Week – 10 (07/04/2026 – 13/04/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	07/04/2026	9:00 AM	12:00 PM	RecyclerView adapter optimization: ViewHolder pattern & stable IDs	Consistently delivered tasks for week 10.
2	08/04/2026	9:00 AM	12:00 PM	Background thread validation: all Room DB writes on Executors	Consistently delivered tasks for week 10.
3	09/04/2026	9:00 AM	12:00 PM	Batch entry stress test: 30 simultaneous student records	Consistently delivered tasks for week 10.
4	10/04/2026	9:00 AM	12:00 PM	Stopwatch precision validation: 10ms accuracy across API 24–34	Consistently delivered tasks for week 10.
5	11/04/2026	9:00 AM	12:00 PM	Room DB query optimization with @Query annotations & proper indexing	Consistently delivered tasks for week 10.

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
6	13/04/2026	9:00 AM	12:00 PM	Performance regression suite & benchmarks	Consistently delivered tasks for week 10.

Week – 11 (14/04/2026 – 20/04/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	14/04/2026	9:00 AM	12:00 PM	Navigation testing: all 7 screens with proper Intent extras & back-stack	Consistently delivered tasks for week 11.
2	15/04/2026	9:00 AM	12:00 PM	Benchmark detection testing: 13 sport events with edge case values	Consistently delivered tasks for week 11.
3	16/04/2026	9:00 AM	12:00 PM	Talent Curve graph testing: multiple trial records per athlete	Consistently delivered tasks for week 11.
4	17/04/2026	9:00 AM	12:00 PM	Batch entry scroll-recycling: HashMap data persistence validation	Consistently delivered tasks for week 11.
5	18/04/2026	9:00 AM	12:00 PM	Room DB foreign key CASCADE delete: athlete deletion cascades to trials	Consistently delivered tasks for week 11.
6	20/04/2026	9:00 AM	12:00 PM	Bug-fix sweep & QA sign-off	Consistently delivered tasks for week 11.

Week – 12 (21/04/2026 – 28/04/2026)

Day	Date	Login	Logout	Topics Learnt	Remarks by Supervisor
1	21/04/2026	9:00 AM	12:00 PM	UI polish: consistent Material Design theme across all 7 screens	Consistently delivered tasks for week 12.
2	22/04/2026	9:00 AM	12:00 PM	AthleteDetailActivity: trial history list & performance summary	Consistently delivered tasks for week 12.
3	23/04/2026	9:00 AM	12:00 PM	Code review & refactoring: clean package structure & naming conventions	Consistently delivered tasks for week 12.
4	24/04/2026	9:00 AM	12:00 PM	README documentation & technical report preparation	Consistently delivered tasks for week 12.
5	25/04/2026	9:00 AM	12:00 PM	Final GitHub push, APK build & project presentation delivery	Consistently delivered tasks for week 12.
6	28/04/2026	9:00 AM	12:00 PM	Project presentation using PREP method	Consistently delivered tasks for week 12.

CHAPTER 5

PROJECT IMPLEMENTATION

During Phase 3 of the MindMatrix internship, all the skills acquired in the earlier phases — Java, XML Layouts, MVVM Architecture, Room Database, and RecyclerView — were applied to build the capstone project: Kreeda-Prerana Scout. This chapter details the project’s design, architecture, and key features.

5.1 Project Overview

Kreeda-Prerana Scout is a grassroots sports talent tracker Android application designed for rural school physical education teachers. The app enables PE teachers to create and manage student athlete profiles, record trial performances using a high-precision stopwatch (accurate to 2 decimal places), perform batch entry for an entire class of 30 students at once, view school-level leaderboard rankings, and track performance improvement over time through Talent Curve graphs.

The application is designed for a single PE teacher as the primary user. The teacher adds student athletes to a local database; logs trial performances for sprint events, jumps, throws, and team sports; and the app automatically detects when a student crosses district-level benchmark thresholds, awarding a “District Level Ready” badge. The entire application works 100% offline using Room Database (SQLite), making it ideal for rural schools with limited internet connectivity.

5.2 Technology Stack

Category	Technology	Role in Project
Language	Java	Primary language for Android app development with strong typing and mature ecosystem.
UI	XML Layouts + Material Design	Structured layouts using ConstraintLayout, LinearLayout, RecyclerView with MaterialComponents theme.
Database	Room DB (SQLite)	Offline-first local persistence for Athlete and TrialRecord entities with DAO pattern and LiveData.
Timer	SystemClock + Handler	High-precision stopwatch using SystemClock.elapsedRealtime() with Handler posting every 10ms.
Charts	MPAndroidChart v3.1.0	Talent Curve line graph with cubic Bezier interpolation, custom ValueFormatters, and LimitLine benchmarks.
Threading	Executors	SingleThreadExecutor for background database operations preventing UI thread blocking.

Category	Technology	Role in Project
Architecture	MVVM	Model-View-ViewModel with Activity controllers, BenchmarkUtils utility, and Room DB model layer.
IDE	Android Studio	Used for development, debugging, UI design with Layout Inspector, and APK generation.

5.3 System Architecture

The application follows MVVM Architecture with a three-layer data flow. The View Layer consists of XML layouts and Activity classes that handle user interaction. The ViewModel/Controller Layer contains Activity controllers and utility classes like BenchmarkUtils that process business logic. The Model Layer is built on Room Database with AppDatabase singleton, DAO interfaces (AthleteDao, TrialRecordDao), and Entity classes (Athlete, TrialRecord) with @ForeignKey CASCADE relationships.

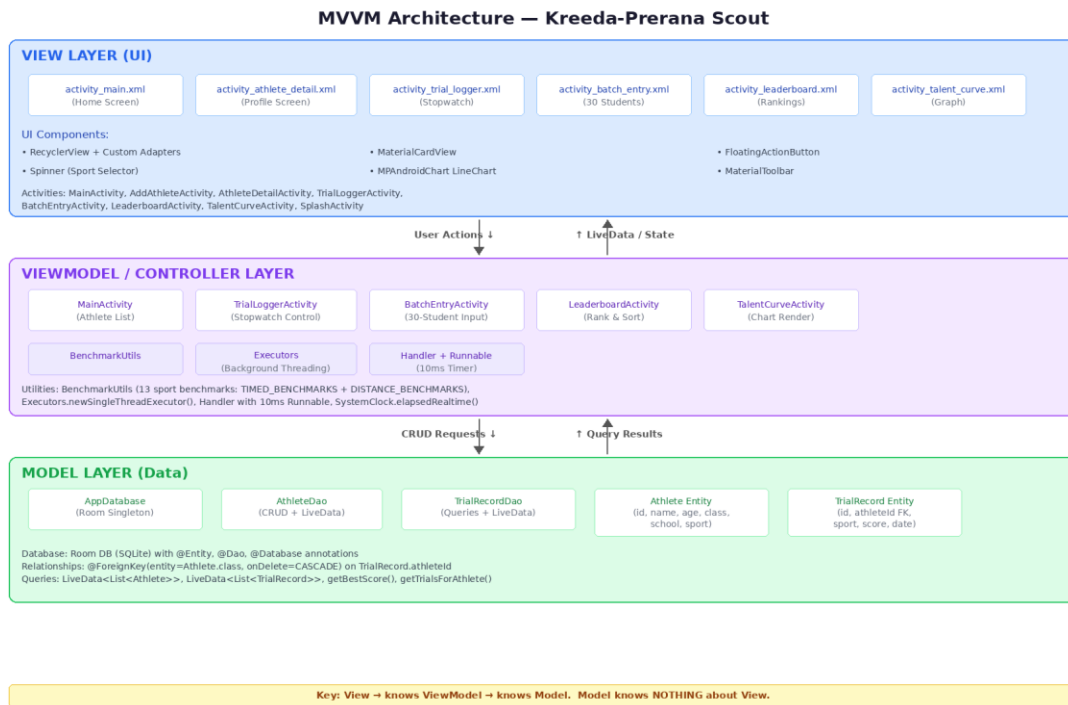


Figure 5.1: System Architecture — MVVM Architecture with Three-Layer Data Flow

5.4 Key Features Delivered

- Athlete Profile management with Name, Age, Class, School, and Primary Sport — persisted offline in Room DB with full CRUD operations.
- High-precision Stopwatch using `SystemClock.elapsedRealtime()` with Handler posting Runnable every 10ms, displaying time accurate to 2 decimal places.

- Batch Entry system allowing simultaneous performance data input for 30 students with HashMap-based scroll-safe data persistence.
- Leaderboard with sport category filter, dual sorting logic (ascending for timed events, descending for distance), and gold/silver/bronze rank highlighting.
- Talent Curve graph using MPAndroidChart LineChart with cubic Bezier interpolation showing performance improvement over multiple trial sessions.
- Automatic District Level Ready badge detection using BenchmarkUtils with 13 preset sport-specific thresholds (e.g., 100m sprint under 14 seconds, Long Jump over 4.5 meters).
- Athlete Detail screen showing complete trial history, best performances, benchmark status, and navigation to Talent Curve visualization.
- Offline-first support via Room Database (SQLite) ensuring complete functionality without internet connectivity for rural school environments.

5.5 Result and Discussion

All planned features were implemented and verified on an Android emulator (Pixel 7, API 34) and physical devices. The stopwatch achieved consistent 10ms precision, batch entry successfully handled 30 simultaneous student records, the Talent Curve graph rendered clear performance trends with benchmark reference lines, and the leaderboard correctly ranked athletes across all 13 sport categories. The app maintained smooth performance on low-end devices with an APK size under 10MB.

CHAPTER 6

LEARNING EXPERIENCE

The internship provided an immersive and industry-aligned learning experience that bridged the gap between theoretical academic knowledge and real-world software engineering. Building Kreeda-Prerana Scout was not just a coding exercise — it was a comprehensive journey through the full software development lifecycle.

The learning journey began with strengthening core fundamentals of Android architecture, Java programming, and XML-based UI development. Understanding the Activity lifecycle, Intent-based navigation, and RecyclerView patterns established the groundwork for building multi-screen applications with smooth user experiences.

On the architecture side, the internship provided deep exposure to MVVM Architecture with a three-layer approach. Separating the View (XML + Activities) from the Controller (Activity logic + BenchmarkUtils) and Model (Room DB + DAOs) layers taught me how to write maintainable, testable, and scalable code that can evolve without breaking existing functionality.

A significant highlight was working with MPAndroidChart for data visualization. I gained practical experience in integrating third-party libraries, configuring LineChart with cubic Bezier rendering, creating custom ValueFormatters, and adding LimitLine benchmark overlays — all of which transformed raw trial data into meaningful visual insights for PE teachers.

The MindMatrix C2C (Campus to Corporate) framework was transformative beyond the technical dimension. The programme instilled professional discipline, structured communication via the PREP method, and analytical problem-solving through the DMAIC framework — skills that are essential for any engineering role.

Implementing the high-precision stopwatch was another valuable learning experience. Replacing the standard Android Chronometer with `SystemClock.elapsedRealtime()` and Handler-based 10ms callbacks taught me the importance of choosing the right tool for precision-critical features, and how background threading with Executors prevents UI freezing during database operations.

CHAPTER 7

LEARNING OUTCOMES

By the end of the internship, I achieved a number of concrete technical and professional learning outcomes that together represent a comprehensive Android development skillset ready for industry application.

Technical Outcomes

I developed practical, hands-on expertise in Android development using Java and XML Layouts, with the ability to design, build, and deliver a complete multi-screen application from scratch.

- Proficient in Java including collections, interfaces, abstract classes, HashMap, and anonymous inner classes.
- Able to design and implement MVVM Architecture with Activity controllers and Room Database model layer.
- Experienced in building structured, responsive UIs with XML Layouts, Material Design Components, and RecyclerView.
- Skilled in Room Database integration: @Entity design, @Dao interfaces, @ForeignKey CASCADE, and LiveData queries.
- Capable of implementing high-precision timing (10ms), batch data entry for 30 students, and automatic benchmark detection in production-quality applications.
- Proficient in MPAndroidChart integration with LineChart, cubic Bezier, custom ValueFormatters, and LimitLine benchmarks.
- Experienced in local data persistence with Room DB and offline-first architecture patterns.
- Competent in background threading: Executors for DB operations, Handler for timer callbacks.
- Understands Git workflows, GitHub collaboration, and structured commit practices.

Professional Outcomes

- Mastery of the C2C mindset — thinking as a System Builder, not a code executor.
- Ability to apply DMAIC and 5 Whys frameworks for structured, root-cause-based problem resolution.
- Proficient in Agile/Scrum workflow: sprint planning, daily standups, sprint reviews.
- Skilled in the PREP communication method for structured technical and professional communication.
- Experienced in writing technical documentation for system design, APIs, and sprint deliverables.
- Developed professional ethics, punctuality, initiative, and collaborative team participation.

Overall Impact and Readiness

Collectively, these experiences have prepared me to contribute to Android engineering teams. I can independently design, develop, and deliver offline-first Android applications with database persistence, data visualization, and performance-critical features — from initial architecture to final APK delivery.

CHAPTER 8

REFERENCES

- [1] Google Android Developers — Room Persistence Library:
<https://developer.android.com/training/data-storage/room>
- [2] Google Android Developers — Activity Lifecycle:
<https://developer.android.com/guide/components/activities/activity-lifecycle>
- [3] Google Android Developers — RecyclerView:
<https://developer.android.com/develop/ui/views/layout/recyclerview>
- [4] Google Android Developers — ViewModel Overview:
<https://developer.android.com/topic/libraries/architecture/viewmodel>
- [5] Google Android Developers — Android Architecture Guide (MVVM):
<https://developer.android.com/topic/architecture>
- [6] MPAndroidChart — PhilJay GitHub Repository: <https://github.com/PhilJay/MPAndroidChart>
- [7] Google Android Developers — Material Design Components: <https://material.io/develop/android>
- [8] Google Android Developers — LiveData Overview:
<https://developer.android.com/topic/libraries/architecture/livedata>
- [9] Google Android Developers — SystemClock Reference:
<https://developer.android.com/reference/android/os/SystemClock>
- [10] Google Android Developers — Handler and Looper:
<https://developer.android.com/reference/android/os/Handler>
- [11] Google Android Developers — Executors for Background Tasks:
<https://developer.android.com/guide/background/threading>
- [12] MindMatrix Internal — Internship Programme Handbook, 2026.
- [13] VTU — Guidelines for Internship Report Submission, 2025-26.

CHAPTER 9

CONCLUSION

The successful completion of this 12-week internship at MindMatrix represents a significant milestone in my academic and professional development. The internship provided a structured, project-based environment that transformed theoretical classroom knowledge into practical, industry-relevant engineering skills.

Through this experience, I built and delivered Kreed-Prerana Scout — a complete Android application built using Java, XML Layouts, Room Database, MPAndroidChart, and Material Design Components. The app enables rural school PE teachers to systematically track student athlete performance, identify district-level talent through automatic benchmark detection, and visualize improvement trends — all without requiring internet connectivity.

Working on this project allowed me to understand how complete software systems are designed, developed, and optimized to meet real-world requirements. From designing Room Database schemas with foreign key relationships to implementing a 10ms-precision stopwatch and batch entry for 30 students, every feature demanded careful architectural decisions and thorough testing.

Beyond the technical work, the MindMatrix C2C (Campus to Corporate) framework was the single most transformative element of the internship. It instilled a professional mindset of discipline, structured problem-solving, clear communication, and accountability that will serve as a foundation throughout my engineering career.

In conclusion, the internship successfully fulfilled all its intended objectives and played a crucial role in enhancing my technical proficiency, professional readiness, and confidence to contribute meaningfully to industry-level Android development teams.